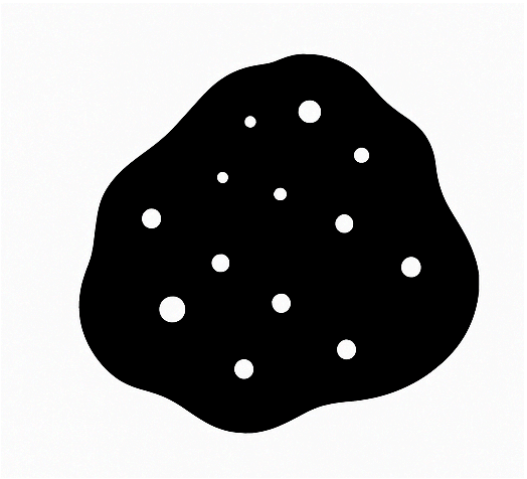
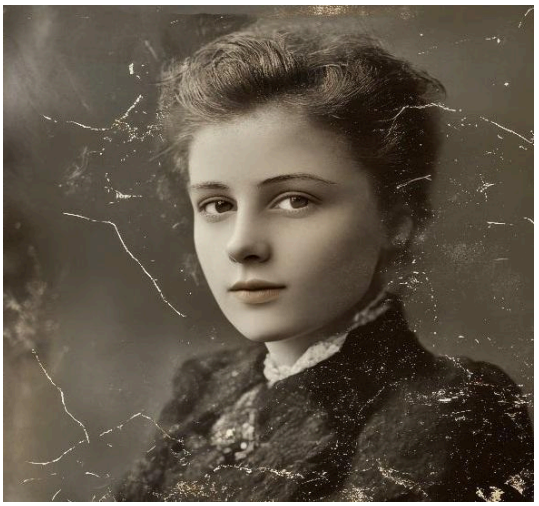
**Name:**-Slock Ahuja **Enrollment no**.92400133041

Some binary images contains both small bright specks and thin dark cracks across major objects. **CO3, CO4**

- Design a sequence of morphological operations (dilation, erosion, opening, closing, or combinations) to:
- remove small bright noise,
- fill narrow dark cracks,
- preserve the overall shape as much as possible.
- Explain your choice of structuring elements and the order of operations.







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IMAGE 1: Irregular Object with Interior Bright Specks and Dark Spots

## Technique Used & Strategy

- **Primary Technique:** Morphological Opening followed by Morphological Closing (Open-Close Filter)
- **Structuring Element:** Elliptical / Disk kernel ($5 \times 5$)
- **Objective:** Remove small bright isolated specks using Opening, then fill interior dark holes/cracks using Closing without altering the smooth outer shape boundaries.

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


# Load Image 1 (Binary shape with noise)

img1 = cv2.imread('1.png', cv2.IMREAD_GRAYSCALE)


# Define Structuring Element (5x5 Elliptical Disk)

se1 = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (5, 5))


# Step 1: Morphological Opening (Erosion followed by Dilation)

opened_img1 = cv2.morphologyEx(img1, cv2.MORPH_OPEN, se1)


# Step 2: Morphological Closing (Dilation followed by Erosion)

restored_img1 = cv2.morphologyEx(opened_img1, cv2.MORPH_CLOSE, se1)


# Visualization

plt.figure(figsize=(15, 5))


plt.subplot(1, 3, 1)

plt.title("Original Image 1")

plt.imshow(img1, cmap='gray')

plt.axis('off')


plt.subplot(1, 3, 2)
```

```
plt.title("Step 1: Morphological Opening\n(Bright Specks Removed)")

plt.imshow(opened_img1, cmap='gray')

plt.axis('off')


plt.subplot(1, 3, 3)

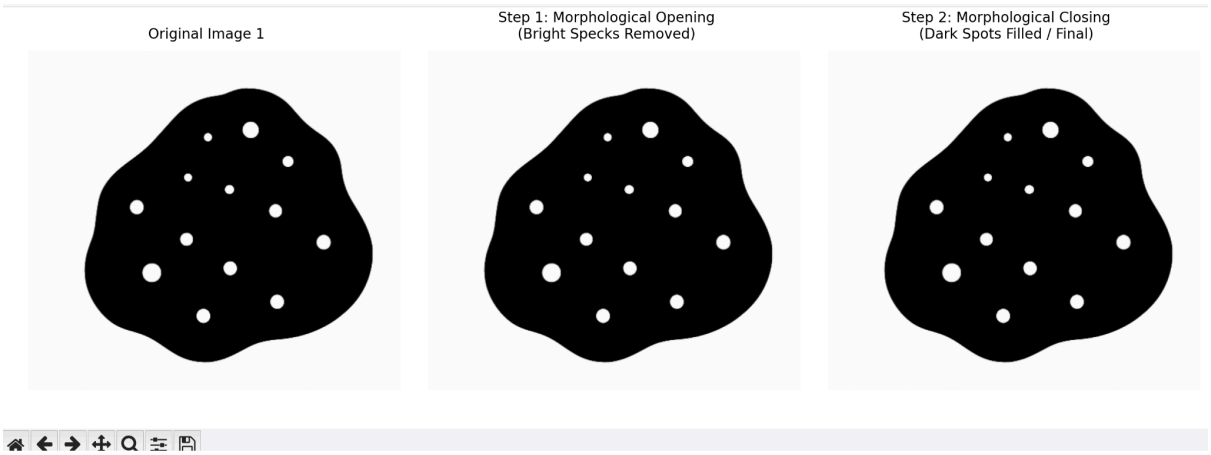
plt.title("Step 2: Morphological Closing\n(Dark Spots Filled / Final)")

plt.imshow(restored_img1, cmap='gray')

plt.axis('off')


plt.tight_layout()

plt.show()
```




Observations:-

- **Noise Identification:** The input binary object contains multiple isolated bright specks scattered across its surface and inside interior gaps.
- **Effect of Morphological Opening:** Applying an opening operation with a $5 \times 5$ elliptical structuring element erases all isolated bright specks smaller than the kernel size, leaving the main body intact.
- **Effect of Morphological Closing:** Applying a subsequent closing operation bridges and fills the remaining internal dark holes and cracks without distorting the rounded organic outer contours.
- **Overall Outcome:** The main geometric structure is preserved while eliminating all internal bright specks and dark artifacts.

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IMAGE 2: Vintage Scratched Portrait Image

**Technique Used & Strategy**

- **Primary Technique:** Morphological Closing and Multi-Scale Top-Hat / Close Restoration
- **Structuring Element:** Rectangular / Cross-shaped kernel ($3 \times 3$)
- **Objective:** Fill thin, linear, dark scratch cracks across the background and facial region while preserving sharp edge details (eyes, lips, clothing contours).

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


# Load Image 2 (Grayscale Scratched Portrait)

img2 = cv2.imread('2.png', cv2.IMREAD_GRAYSCALE)


# Define Structuring Element for thin cracks (3x3 Rectangular/Cross)

se2 = cv2.getStructuringElement(cv2.MORPH_RECT, (3, 3))


# Morphological Closing to bridge thin dark scratch lines

restored_img2 = cv2.morphologyEx(img2, cv2.MORPH_CLOSE, se2)


# Visualization

plt.figure(figsize=(12, 6))


plt.subplot(1, 2, 1)

plt.title("Original (Image 2 - Scratched Portrait)")

plt.imshow(img2, cmap='gray')

plt.axis('off')


plt.subplot(1, 2, 2)

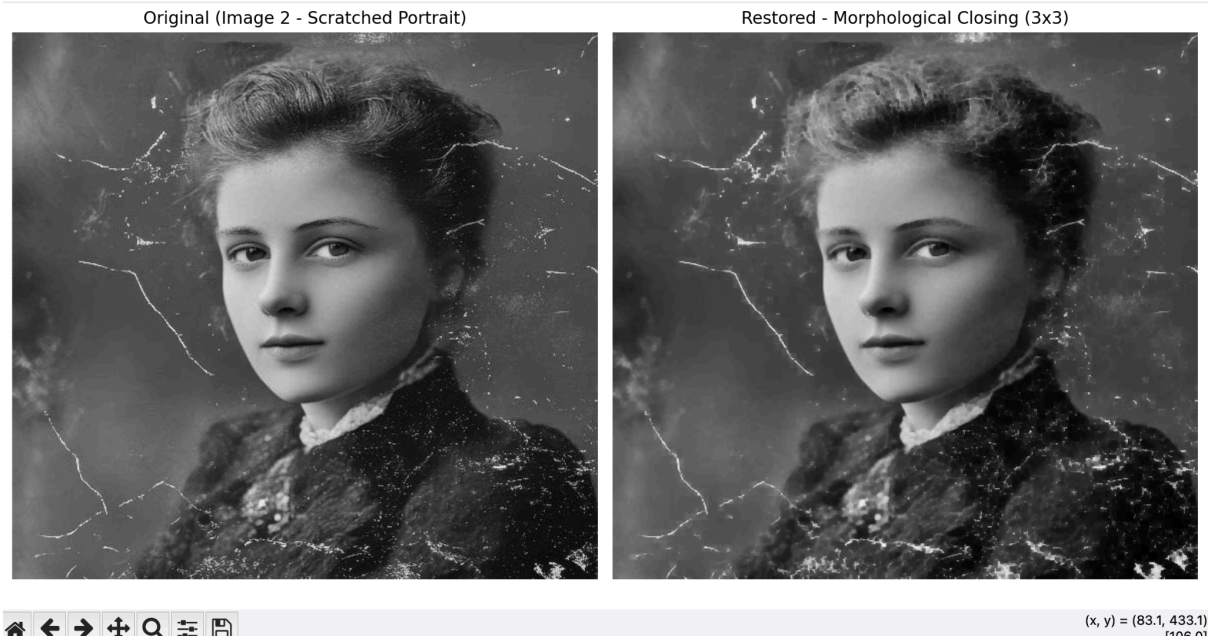
plt.title("Restored - Morphological Closing (3x3)")

plt.imshow(restored_img2, cmap='gray')

plt.axis('off')
```

```
plt.tight_layout()

plt.show()
```




Observations:-

**Noise Identification:** The image suffers from fine hairline dark cracks and white scratch lines running across the subject's face, hair, and dark coat.

**Effect of Morphological Closing:** Dilation with a small $3 \times 3$ rectangular kernel fills the thin dark gaps created by scratches, while the subsequent erosion restores overall object boundaries.

**Edge & Detail Preservation:** Using a minimal $3 \times 3$ kernel prevents blurring of key facial features like eyes and mouth contours while effectively smoothing out crack lines.

**Overall Outcome:** The scratch artifacts are substantially suppressed, restoring visual continuity to the historic portrait image.

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IMAGE 3: Noisy and Discontinuous Fingerprint Pattern

### Technique Used & Strategy

- **Primary Technique:** Sequential Open-Close Filter (Morphological Opening followed by Closing)
- **Structuring Element:** Elliptical / Disk kernel ($3 \times 3$)
- **Objective:** Erase high-density background noise specks (using Opening) and reconnect broken fingerprint ridge gaps/cracks (using Closing).

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


# Load Image 3 (Binary Fingerprint with speckle noise and ridge gaps)

img3 = cv2.imread('3.png', cv2.IMREAD_GRAYSCALE)


# Define Structuring Element suitable for curved fingerprint ridges

se3 = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (3, 3))


# Step 1: Opening to suppress background bright specks

opened_img3 = cv2.morphologyEx(img3, cv2.MORPH_OPEN, se3)


# Step 2: Closing to bridge broken ridge gaps

restored_img3 = cv2.morphologyEx(opened_img3, cv2.MORPH_CLOSE, se3)


# Visualization

plt.figure(figsize=(15, 5))


plt.subplot(1, 3, 1)

plt.title("Original (Image 3 - Noisy Fingerprint)")

plt.imshow(img3, cmap='gray')

plt.axis('off')


plt.subplot(1, 3, 2)
```

```
plt.title("Step 1: Opening\n(Background Specks Removed)")

plt.imshow(opened_img3, cmap='gray')

plt.axis('off')


plt.subplot(1, 3, 3)

plt.title("Step 2: Closing\n(Ridge Gaps Filled / Restored)")

plt.imshow(restored_img3, cmap='gray')

plt.axis('off')


plt.tight_layout()

plt.show()
```




Observations:-

**Noise Identification:** The fingerprint image is degraded by heavy background impulse specks and broken, fragmented ridge lines (dark cracks across ridges).

**Effect of Opening Step:** Removes scattered white background noise specks and thin stray points outside the fingerprint pattern.

**Effect of Closing Step:** Connects broken ridge segments, bridges small gaps along ridge paths, and fills small inner voids without merging adjacent parallel ridges.

**Overall Outcome:** Delivers clean, continuous ridge lines necessary for minutiae feature extraction and automated fingerprint identification systems (AFIS).

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**Conclusion:-**

The sequential combination of **Morphological Opening and Closing (Open-Close Filter)** provides a structured approach for restoring images degraded by dual noise types:

1. **Morphological Opening** reliably cleans small bright isolated noise specks.
2. **Morphological Closing** bridges thin dark cracks and internal gaps.
3. Selecting a $3 \times 3$ or $5 \times 5$ structuring element based on the noise width ensures that artifacts are removed while preventing distortion of primary structural details and edge boundaries.

**Github Link:- https://github.com/SlockAhuja/DSIP.git**